Contribution to the discussion of “Safe Testing” by Grünwald, de Heide, and Koolen

Neil Dey, Ryan Martin, and Jonathan P Williams
Department of Statistics
North Carolina State University
Raleigh, North Carolina, USA

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Congratulations to Grünwald, de Heide, and Koolen (GHK) for their contribution to the exciting and rapidly-growing literature on safe/anytime-valid inference. GHK's proposal focused on problems in which a statistical model is available that determines the quantity of interest—the model parameter—and provides a likelihood function that drives the e-value construction. When data are observed and analyzed sequentially, however, the data analyst typically cannot say with any certainty what statistical model might be appropriate. Our comments below focus on what can be done along the lines of safe/anytime-valid inference without a correctly-specified statistical model.

Two recent developments deserve mention. First, Park et al. (2023) develops a version of the *universal inference* framework (Wasserman et al. 2020) that accommodates model misspecification but is not designed for the online setting. Second, the nonparametric approach in Waudby-Smith and Ramdas (2024) offers anytime-valid inference, but focuses exclusively on the mean of the data-generating process.

In Dey et al. (2023), we take a different approach, starting from the basic question: What is the inferential target? From classical M-estimation to modern machine learning, the inferential target is often the solution to a suitable optimization problem. These solutions can be cast as risk minimizers: Given a loss function $\ell_\theta(z)$, the inferential target is $\theta^\star = \arg\min_\theta R(\theta)$, with $R(\theta) = \int \ell_\theta(z)\, P(dz)$ the risk. We generalize the universal inference framework by replacing the statistical model's negative log-likelihood with a multiple of the empirical risk, $R_n(\theta) = n^{-1} \sum_{i=1}^n \ell_\theta(Z_i)$, and prove that it offers safe, anytime-valid inference on risk minimizers.

Circling back to GHK's focus on growth rate optimality, we show that our generalized universal e-process for testing $H_0 : \theta^\star \in \Theta_0$ versus $H_1 : \theta^\star \in \Theta_1$, has (asymptotic) log-growth rate proportional to

$$n \Big\{ \inf_{\theta \in \Theta_0} R(\theta) - \inf_{\theta \in \Theta_1} R(\theta) \Big\} + o(n), \quad P\text{-almost surely.} \tag{1}$$

The above claim is analogous to Theorem 2 in Dixit and Martin (2023). Our conjecture is that (1) gives the “optimal” asymptotic growth rate achievable by an e-process relative to the given learning problem (P, ℓ). If our conjecture is true, then given the similarities between GHK's growth rate and (1), one could approach this more general problem similarly to GHK, by *defining* an e-process based on a growth rate optimality criterion, and we expect its logarithm would be proportional to an empirical risk difference. Further, the corresponding "growth rate optimal" e-process would have to be similar to what we proposed in Dey et al. (2023).

References

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